Happiest Health Digital Twin Challenge 2026

Glucose Excursion Twin

A physiology-guided Digital Twin that warns a care team about a glucose excursion two hours before it happens, and says how sure it is.

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Research prototype. Not a medical device.

India’s diabetes care is reactive, and CGM data goes unwatched

101 M

adults living with diabetes

136 M

more with prediabetes

HbA1c every few months misses what happens in between.

CGMs make the in-between visible, but no clinic team can watch hundreds of continuous traces.

Source: ICMR-INDIAB-17, Anjana RM et al., Lancet Diabetes Endocrinol 2023

One twin per patient, one queue for the care team

Who uses it

A diabetes educator or physician monitoring a panel of patients who wear CGMs.

What it does

Updates each patient’s virtual state every 5 minutes, forecasts 2 hours ahead and ranks patients by risk.

The event

Any CGM reading ≥ 180 mg/dL in the next 120 minutes, predicted while glucose is still below it.

180 mg/dL is the upper limit of the international time-in-range target. Here it is a configurable research setting, not a treatment target.

Open, anonymised and synthetic data only

Source	People	Role	Licence
CGMacros (PhysioNet)	45 adults: 15 healthy, 16 prediabetes, 14 T2D	Train and test	CC BY-NC-SA 4.0
ShanghaiT2DM (Figshare)	100 adults with T2D	External test only	None attached
Synthetic patients	6 generated personas	Public demo	Ours

A data-quality catch. The CGMacros release fills Dexcom readings onto a 1-minute grid by straight-line interpolation, which leaks future readings into any model that uses those rows. We recovered the 125,819 real 5-minute readings and test for it.

A grey-box twin: physiology first, machine learning for what it misses

$$\frac{dG}{dt} = -p\,(G - B) + R$$

G sensor glucose, observed every 5 min

B personal baseline the patient returns to

R unexplained inflow: meals, liver, stress

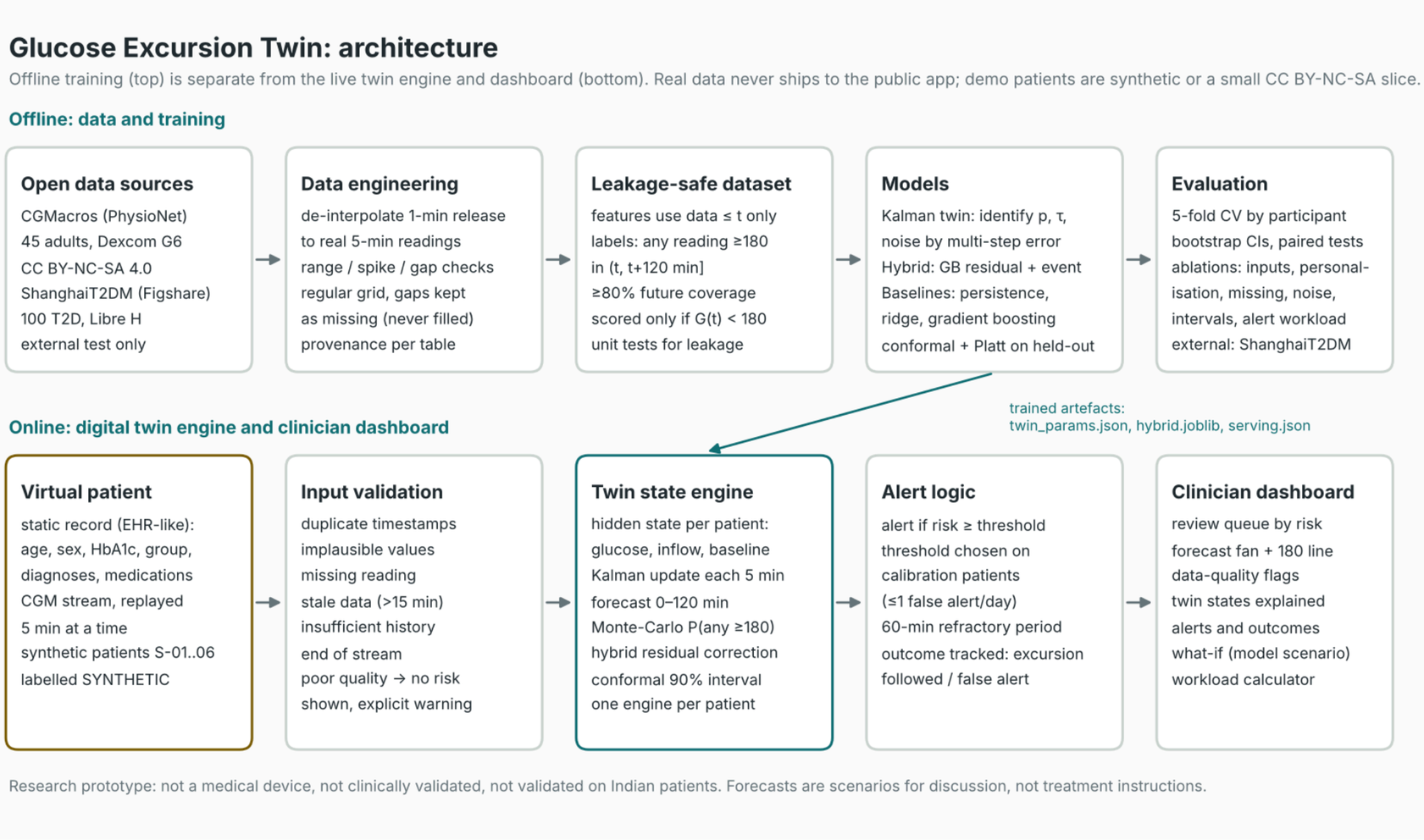
A Kalman filter updates the hidden states with each reading and widens its uncertainty when readings are missing.

Then the hybrid

Gradient boosting learns the twin’s residual error and the 2-hour excursion risk from its states, recent glucose patterns and the profile.

Intervals are 90% conformal, calibrated on held-out patients, separately for each diabetes group.

Training stays offline; the live twin runs per patient



We tested the twin against simpler models, on people it never saw

Split by person

5-fold cross-validation over 45 participants. No one appears in both training and testing.

No tuning on the test

Intervals, probabilities and the alert threshold are set on separate calibration patients.

Five rivals, same data

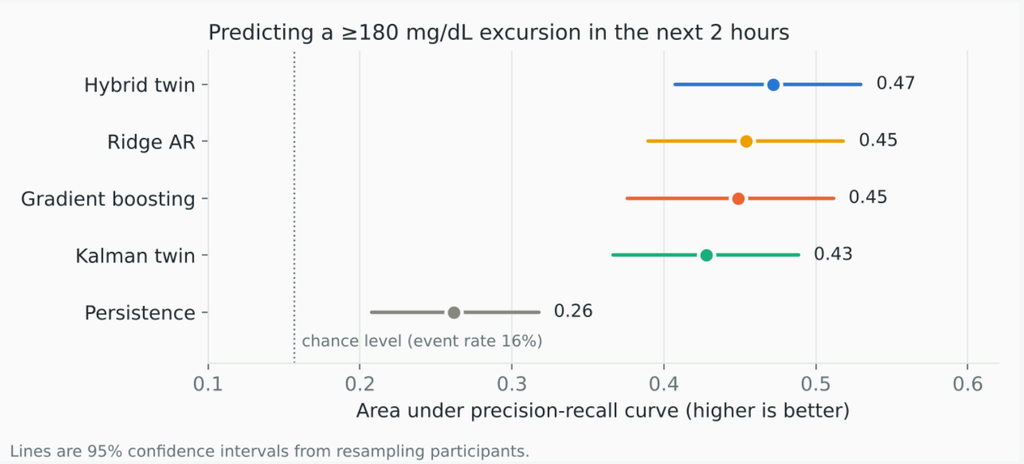
Persistence, ridge regression, gradient boosting, the pure Kalman twin and the hybrid.

Stress tests

Missing readings, sensor noise, personalisation, and 100 external patients from Shanghai.

The hybrid twin scores best on every event metric

Model	MAE 60	MAE 120	AUROC	AUPRC
Persistence	19.4	26.8	0.62	0.26
Ridge AR	17.4	21.5	0.79	0.45
Gradient boosting	17.3	21.5	0.80	0.45
Kalman twin	17.6	21.4	0.78	0.43
Hybrid twin	17.2	21.1	0.81	0.47



MAE in mg/dL at 60 and 120 minutes. AUROC and AUPRC for predicting a reading ≥ 180 mg/dL in the next 2 hours; chance AUPRC is 0.16. 45 participants, 5,427 excursion labels.

But its accuracy gain over plain gradient boosting is not statistically significant.

−0.34 mg/dL

mean 2-hour error difference per person, 95% CI −0.93 to +0.24

28 of 45

people where the hybrid was more accurate at 2 hours

Ridge regression nearly matches gradient boosting, and age, sex and HbA1c add nothing beyond the glucose history. The twin’s value lies elsewhere.

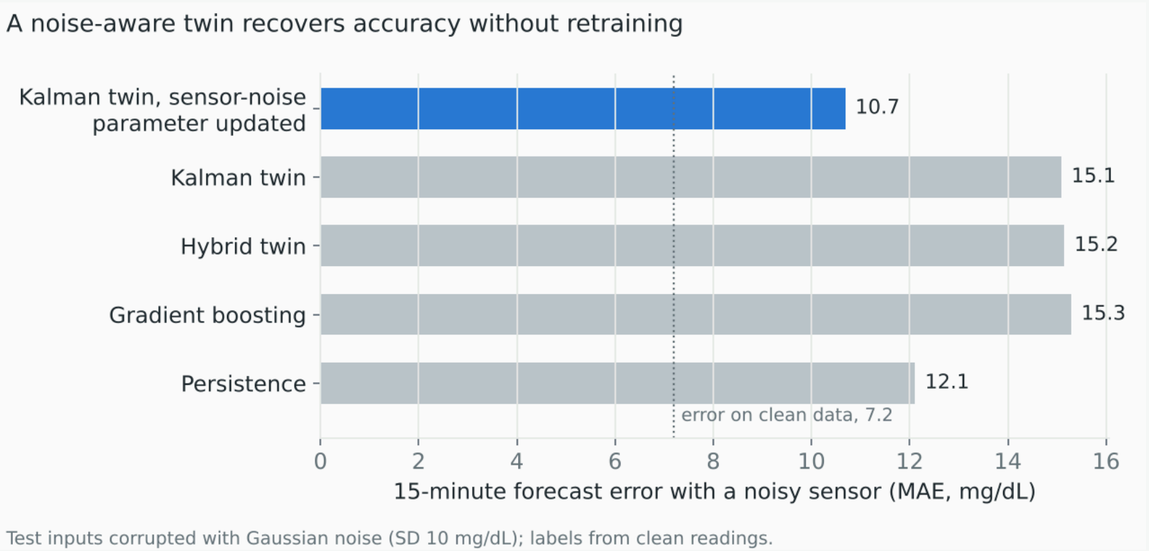
Where the twin’s structure pays off: messy real-world data

30% of readings missing

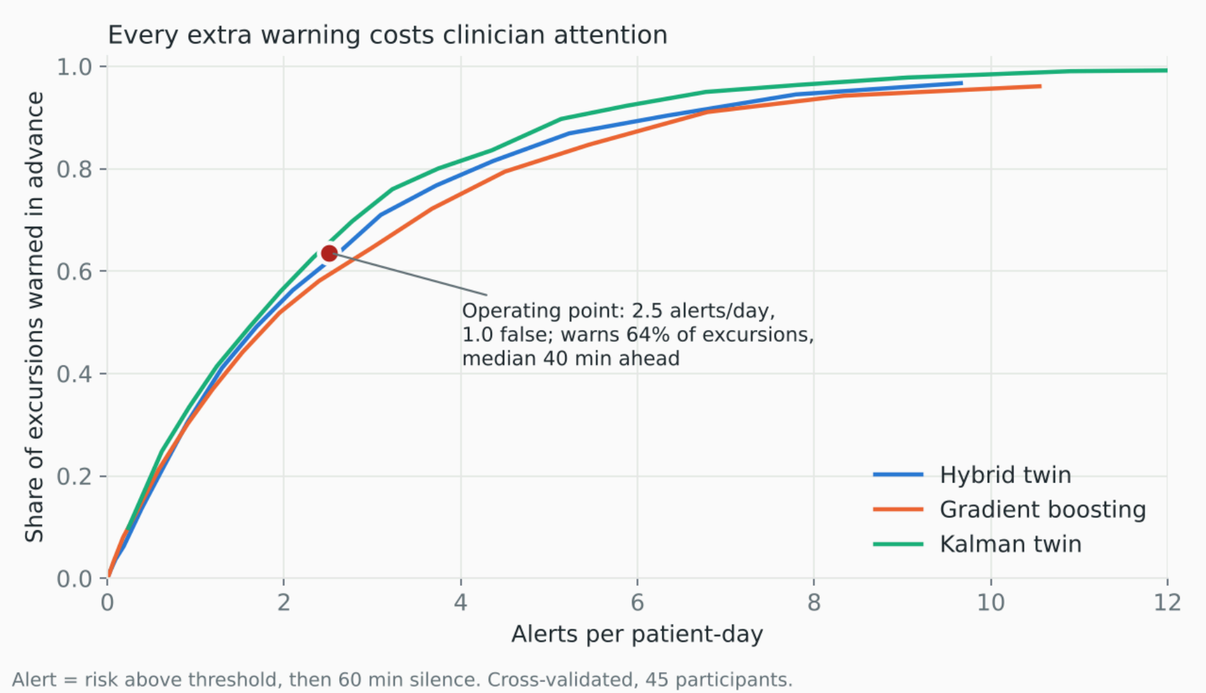
Twin: 60-minute error 17.6 → 17.6 mg/dL, unchanged. Ridge regression: 17.4 → 18.8.

A noisy sensor

Changing one physical parameter, the sensor noise, cuts 15-minute error from 15.1 to 10.7 mg/dL. No retraining.



Every warning costs clinician time, so we set an alert budget



Alert = risk above threshold, then 60 min silence. Cross-validated, 45 participants.

64%

of excursions warned in advance

40 min

median warning time

1.0

false alert per patient-day

On 100 new patients from Shanghai, plain gradient boosting transferred best

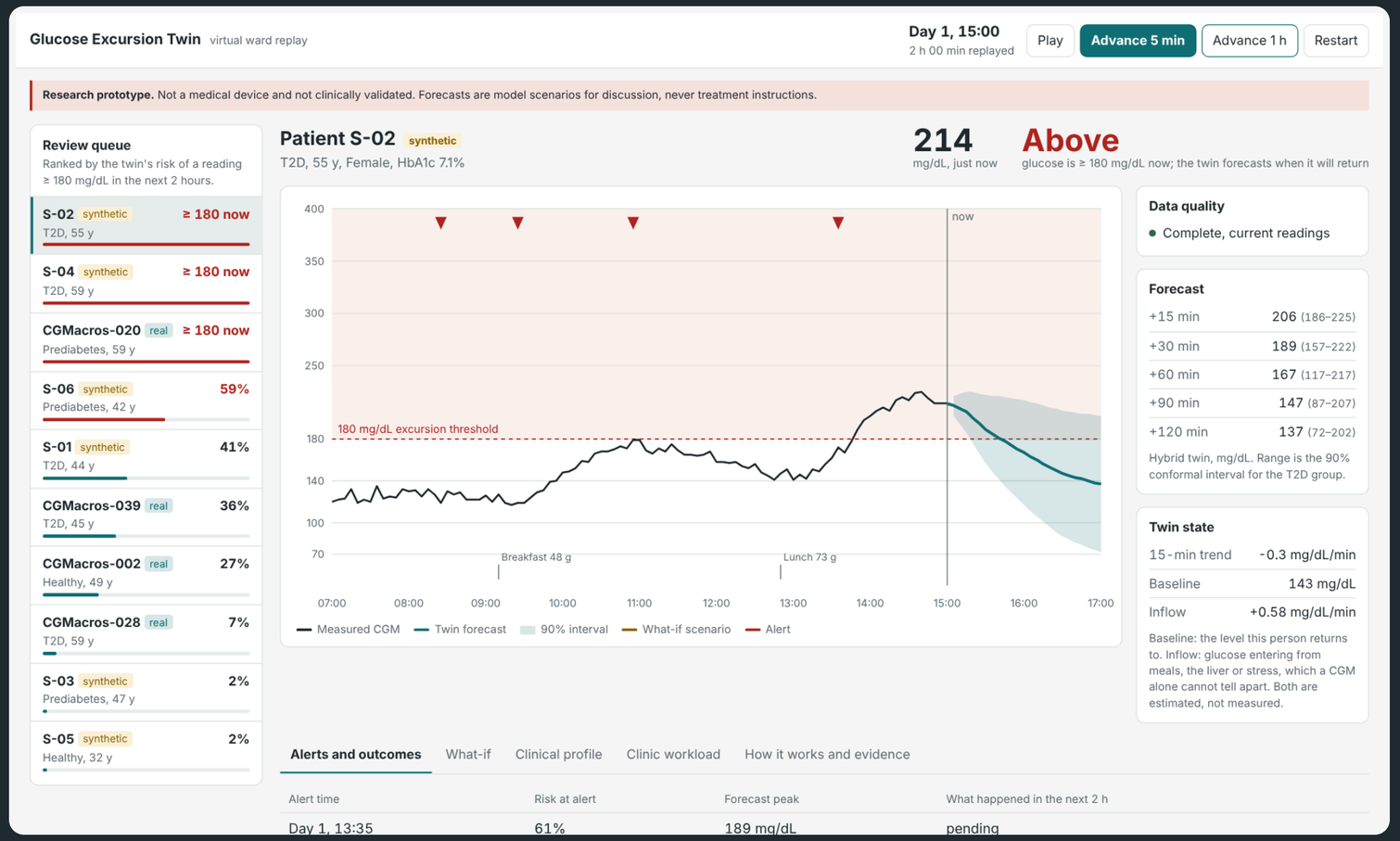
Model	MAE 60	MAE 120	AUROC	AUPRC
Persistence	20.1	31.1	0.75	0.40
Ridge AR	19.9	28.2	0.80	0.43
Gradient boosting	19.6	27.1	0.80	0.45
Kalman twin	20.7	29.3	0.75	0.35
Hybrid twin	20.6	28.7	0.79	0.42

What it means for India

Parameters learned on 45 Californians did not carry over to Chinese hospital patients on a different sensor. Any use in Indian patients needs local recalibration first.

Glucose-only models, 15-minute readings, trained on CGMacros and never pooled with the Shanghai data. Event rate 15%.

The prototype: a virtual ward the care team can click through



Review queue ranked by 2-hour risk

Forecast fan with a 90% interval

Refuses to guess when data are stale or sparse

Alerts with outcomes, what-if meals, clinic workload

For a clinic, the alert policy is the business decision

Policy, 200 patients	Alerts / day	False / day	Review hours	Excursions warned
Warn most (≤ 1 false per patient)	500	200	25	64%
Balanced (risk ≥ 50%)	420	160	21	56%
Quiet (risk ≥ 70%)	128	34	6	22%

Route by tier

High risk to the nurse now; the rest batched into a daily review.

Start where risk is high

T2D patients have nearly 4× the excursion rate of healthy people, and more true alerts.

Prove it first

A silent-mode pilot measures real alert value before any patient is contacted.

Illustrative: rates from cross-validated replay on CGMacros, scaled to 200 patients, 3 minutes per review. Not observed in a clinic.

What it cannot do yet, and what comes next

Limits

- 45 training participants, 14 with T2D, one US site
- No meal, insulin or medication inputs in this run
- Intervals cover 87–88% against a 90% target
- Not validated on Indian patients

Next steps

- Add CGMacros meals and activity to the twin
- Retrain the hybrid on personalised twin states
- Recalibrate on an Indian CGM cohort
- Silent-mode pilot with a diabetes clinic

A twin that knows when it is unsure is one a clinician can trust.

Open data, tested on people it never saw, honest about where it wins.

Code and report

github.com/Mav-e-r-ick/MB25026_IITMandi

Clickable prototype

mav-e-r-ick.github.io/MB25026_IITMandi/prototype

Licence

Code MIT; data and models CC BY-NC-SA 4.0